Algorithm 1 Lifted SCSI EM with Conditional Flow Map Parameterization

Require: Data sampler p_{data}; forward corruption $\mathcal{F}(x) = x + \sigma\varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$; outer iterations K; inner iterations N; batch size B;

Initialize flow-map parameters θ

$\phi \leftarrow \theta$ ▷ frozen generator (copy of θ, no grad)

for $k = 1, \ldots, K$ **do** ▷ outer EM iterations

  for $i = 1, \ldots, N$ **do** ▷ inner gradient steps

    Sample $z, z' \sim \mathcal{N}(0, I)^B$; $s, t, t' \sim \mathcal{U}[0, 1]^B$

    Sample $y_{\mathrm{real}} \sim p_{\mathrm{obs}}^B$

    E-step (no grad).

      $x_{\mathrm{fake}} = X^{\phi}_{0\to 1}(z \mid y_{\mathrm{real}})$ *(single forward pass through frozen ϕ)*

      $y_{\mathrm{fake}} \leftarrow \mathcal{F}(x_{\mathrm{fake}})$

      Compute $I_{t'} \leftarrow (1 - t')\, z' + t'\, x_{\mathrm{fake}}$ and $\dot{I}_{t'} \leftarrow x_{\mathrm{fake}} - z'$

    M-step. Compute the loss and update θ:

      $\tilde{x} := X^{\theta}_{t\to s}(I_t \mid y_{\mathrm{fake}})$

      $$\mathcal{L}_{\mathrm{cm}}(\theta) = \left\| \frac{d}{dt} X^{\theta}_{s\to t}(\tilde{x} \mid y_{\mathrm{fake}}) - \dot{I}_t \right\|^2 + \left\| X^{\theta}_{s\to t}(\tilde{x} \mid y_{\mathrm{fake}}) - I_t \right\|^2.$$

      $\theta \leftarrow \theta - \eta_\theta\, \nabla_\theta\, \mathcal{L}_{\mathrm{cm}}$

  end for

  $\phi \leftarrow \theta$ ▷ update frozen generator after each outer iteration

end for

return θ ▷ trained conditional flow map